

Kairav Bhatia

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EDUCATION

Vivekananda Institute Of Professional Studies, GGSIPU

New Delhi, India

B.Tech in Computer Science and Artificial Intelligence; CGPA: 7.85/10.0

July 2022 – July 2026 (expected)

DAV Public School CBSE; Class X: 81% (2020), Class XII: 83% (2022)

SKILLS

Languages: Python, C++, Go, JavaScript

Tools: Node.js, Express.js, Flask, FastAPI, RESTful APIs, MySQL, MongoDB, React.js, Git, Linux, Bash, Nginx, Docker, CI/CD (GitHub Actions, Jenkins)

Coursework: Data Structures and Algorithms, Object Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, System Design

EXPERIENCE

- **Secure Elements** — *Software Engineering Intern, Remote* *March 2026 – Present*
 - Built and maintained Jenkins CI/CD pipelines for Yocto-based embedded systems, internal applications, and AI models, enabling automated build, testing, and deployment for NXP S32G platforms.
 - Engineered an automotive cybersecurity log simulator with AUTOSAR-style IDS, OTA, and UDS log generation, syslog conversion, and vehicle theft attack emulation to support MSOC workflows and automotive security analysis.
- **Devnovate.co** — *Software Engineering Intern, Remote* *May 2025 – January 2026*
 - Deployed and managed a full-stack application on a hardened Debian server using Nginx and PM2, implementing SSH key authentication, restricted firewall rules, and user permissions to improve server security.
 - Supported **1,000+ users** in the first week with under 300ms response times, scaling to 100k+ total users.
- **Self Employed** — *Freelance Software Engineer, Remote* *May 2024 – May 2025*
 - Shipped software products to **10+ clients globally**, owning projects end-to-end from design to deployment. Built secure web apps, Linux servers, and APIs, delivering stable and maintainable production systems.

PROJECTS

- **Scalable URL Shortening Service** — *Go, APIs, Concurrency, Rate Limiting, Caching, Load Testing* [Link](#)
 - Built a backend URL shortening service in Go with clean REST APIs and collision-safe short code generation.
 - Implemented concurrency-safe request handling with token-bucket rate limiting, in-memory caching for hot URLs, persistent storage, and health/readiness endpoints to improve performance and reliability.
 - Stress-tested the service under concurrent load to measure throughput and latency, identified bottlenecks, and optimized request handling based on observed performance metrics.
- **Linux Command-Line Assistant Tool** — *Go, Bash, Linux* [Link](#)
 - Built a Go-based Linux CLI tool that converts natural language queries into shell commands, provides clear explanations, supports flag-based controls, monitoring mode, and maintains command history logging.
 - Implemented safety-first execution logic including dry-run mode, explicit user confirmation, command validation, and warnings for potentially destructive commands to ensure controlled and reliable interaction with Unix systems.
- **Virtual Banking System** — *MongoDB, Node.js, Express.js, JavaScript* [Link](#)
 - Built a banking backend system with user account creation, transaction history, fund transfers, and role-based dashboards for users and administrators, executed via RESTful APIs and structured data models.

COMPETITIVE PROGRAMMING

- **Solved 700+** algorithmic problems across [Leetcode](#), [Codeforces](#), [TakeUForward](#), and other online platforms.
- Achieved competitive programming ratings: Leetcode (max 1652), Codeforces (max 1081).

ACHIEVEMENTS

- Completed **Harvard CS50x** (Introduction to Computer Science), including 10 problem sets and one final project.
- **1st place** in web technologies quiz competition at VIPS-TC, Pitampura.
- **1st place** in debugging competition at Amity University, Noida.
- Reached highest competitive league on Chess.com.
- Leveraged Linux as a daily driver since 2021 for development, experimentation, and community contribution.
- Maintained Github and Stack Overflow profiles since 2018 and 2019 respectively.